

Software

Year 9 Computing Technology

Learning Intention

- Investigate how hardware and software are used to manage, control and secure the movement of, and access to, data in networked digital systems



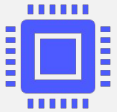
Success Criteria



I can define software and describe its purpose.



I can outline a broad range of software applications.



I can differentiate between application and operating system software.



I can explain the link between hardware and software performance.

What Is Software?



Software: the programs and other operating information used by a computer





Enables computers and devices to perform tasks



Written in **programming languages**, graphical icons, or English-like expressions



Converts input data into useful output or stored information

Application Software





A type of computer program designed to help end-users perform specific personal, educational, or professional tasks



Examples: word processors, graphics editors, web browsers



Mobile apps: small programs for focused tasks, easy to create using user-friendly software



Embedded software: controls single-purpose devices (ATMs, traffic lights, microwaves)



Helps users create, edit, display, or transmit data

Communication Software

Internet browsers: access websites, web apps (online banking, shopping)

Instant messaging: real-time text communication (e.g. Messenger)

File transfer programs: upload/download files (to web servers, wikis)

VoIP software: internet phone calls (e.g., Skype)

Videoconferencing: video/audio calls, chat, multi-user document editing (Zoom, MS Teams)

Example: Emergency Mobile Alert system – sends alerts via mobile towers

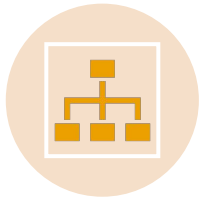
Business Software



Word processing and email essential for modern business



Intranets/knowledge management: share info across teams



Project management: schedule tasks and resources



Resource planning: manage staff, finances, and operations



Data analysis tools: spreadsheets to specialised analytical software



Supports decision-making and efficiency in large organisations

Software for Schools



Standard software installed for all students



Elective computing students use advanced and creative tools



Administrative software: reporting, databases, submissions to authorities



Database-driven systems for student records and reporting

Science and Research Software



Specialised software for data analysis and new knowledge discovery



Uses **AI and expert systems** to support decisions



Current applications: fingerprint matching, cancer diagnosis



Supports complex computations and pattern recognition

Applications for Mobile Devices



Mobile apps (“apps”) for smartphones and tablets



Games, productivity tools, educational software



User-friendly software enables anyone to create apps without advanced programming



Apps allow portability, instant updates, and easy distribution

Operating Systems

A collection of software that manages a computer's hardware and applications by allocating resources, including memory, CPU, input/output devices and file storage



Examples: **Windows, macOS**

Provides startup interface (desktop/logon screen)

Manages orderly shutdown of programs and hardware

Essential for system stability and application compatibility

System Requirements and Performance



Software needs minimum hardware specs: CPU, RAM, GPU, storage



High-demand programs (graphics, CAD) need powerful hardware (video cards)



RAM usage: OS + running programs + program data



Virtual memory: hard disk used if RAM is full



Monitor specs: size, resolution, refresh rate affect performance



Storage: secondary or cloud storage for programs and document